Effect size and practical importance

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Reproducible AI Research Study

THE CONNECTED WEEK

Ask how much changed, in which units and against which prespecified criterion.

Teach useful magnitudes with clear mean-versus-individual and descriptive-versus-inferential limits.

Keep one habit: Model A–B gain convention → apply inclusive mean target → contrast absolute/relative error change → assess new threshold → retain quality/resource/uncertainty gaps.

MAIN LESSON · SECTION 1

Prepare the connected research investigation

> Keep modelling, shared practice and independent evidence distinct.

Target and main teaching plan

Primary target: Explain the **mechanism** and terminology of **effect size** and practical importance

Session · Sequence and minutes · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Sequence and minutes:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Use Your mission, Notice the evidence, See how it works, Words that unlock the idea and Follow a worked example; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Use Find and repair a mistake, Practise together and Run the investigation, with workbook W2 and Individual reasoning check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use Run the investigation, Build your understanding, Connect your project and Show what you know. Reserve further time for the full task set. Use a code example only where one is actually supplied.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building.

Repeated copies of a case share one recorded attempt; they are not new **evidence**.

Preparation and prerequisite

Lesson and workbook; paper and pencil

Calculator and optional local Python 3 for arithmetic; no live API or private data

Prepare the fresh retry separately and withhold its solution until after collection

A result depends on the **input**s, procedure and conditions that produced it. Preserve data roles and distinguish observed outcomes from **prediction**s or unmeasured conditions.

If the prior concept cannot be explained, use a short retrieval activity before this chapter.

Opening retrieval

Without opening the old lesson, explain one key idea from Uncertainty intervals. Give an example and a mistake that the idea helps you avoid.

Look for: Explain the **mechanism** and terminology of uncertainty intervals

Reference example, not the only acceptable wording: Teacher explanation: An uncertainty interval needs a sampling model and assumptions.

A confidence level refers to the repeated-sampling method, not a probability attached to a fixed unknown parameter after observing data.

For this constructed exercise assume independent normal measurements with known population standard deviation 2. A 95% mean interval uses supplied normal critical value 1.96:mean plus/minus1.96*sigma/sqrt(n).

Unknown sigma requires a different justified method.

Without opening the old lesson, explain one key idea from Confounders and controls. Give an example and a mistake that the idea helps you avoid.

Look for: Explain the **mechanism** and terminology of confounders and controls

Reference example, not the only acceptable wording: Teacher explanation: A confounder changes alongside the intended factor and can offer another explanation. Controls aim to isolate the proposed **mechanism**.

For a toy comparison vary only retrieval method while holding corpus, queries, cutoff and relevance rubric fixed. Randomise or counterbalance order when order effects matter.

Without opening the old lesson, explain one key idea from Hypotheses and measurable outcomes. Give an example and a mistake that the idea helps you avoid.

Look for: Explain the **mechanism** and terminology of hypotheses and measurable outcomes

Reference example, not the only acceptable wording: Teacher explanation: A hypothesis predicts a measurable comparison under stated conditions. Declare primary outcome and direction before inspecting results.

Toy hypothesis:method B improves supported-answer rate over baseline A on a reserved paired set. Define support rubric and failure treatment; record secondary outcomes separately.

Pictured end goal

Explain **effect size** and practical importance using the records and boundaries of this fictional investigation.

<details>
<summary>Earlier wording, alternatives and picture transcripts</summary>

Teach the learning cycle

Retrieval answer guidance

Uncertainty intervals

Confounders and controls

Hypotheses and measurable outcomes

Compact alternative: two 45-minute sessions for the core explanation and W1–W5 only. The main plan uses three sessions; schedule further time for the complete investigation, project, visual journey and any code examples. **Effect size** describes magnitude on a stated scale. Practical importance depends on task requirements, not merely a nonzero difference.

Use the supplied constructed scenarios to make the **mechanism** inspectable. Collect W4 before sharing solutions.

Students should name intermediate **evidence** and limitations; classroom pacing and learning effectiveness remain unvalidated.

Outcomes

Explain the **mechanism** and terminology of **effect size** and practical importance

Apply the stated procedure to a new case with a traceable result

Identify an assumption or limitation and design a revealing follow-up
</details>

Preparation

Prerequisite answer

End goal for the pictorial case

This added worked case illustrates the week's main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

MAIN LESSON · SECTION 2

Teach descriptive magnitude and task-specific usefulness without inferential overclaims

> Keep time, errors-per-thousand, percentage points and relative changes separate.

Definitions and complete effect/criterion contract

Effect size describes magnitude on a stated scale. Practical importance depends on task requirements, not merely a nonzero difference.

For paired response durations compute baseline minus candidate per case and average the differences. Positive means faster candidate. Compare with a prespecified minimum useful gain.

Effect/decision item	Complete supplied-exercise rule
Main paired cases	Q1: 10→8; Q2: 12→9; Q3: 8→7 seconds, baseline→candidate
Gain definition	Baseline seconds – candidate seconds; positive is faster candidate
Summary	Mean of the three paired gains, in seconds
Prespecified useful-gain rule	Mean gain at least 2 seconds; equality meets the rule
Mean versus individual cases	Rule applies to the mean, not a requirement that every case gains 2 seconds

Effect/decision item	Complete supplied-exercise rule
Separate error-count picture	A 20 errors/1000 tasks; B 15/1000 initially, changed B 8/1000
Picture useful-change rule	At least 10 fewer errors per 1000; fictional task criterion, not a universal threshold
Absolute and relative effects	Keep errors-per 1000 and percentage points separate from relative reduction
Evidence limits	No uncertainty, significance, causal effect or complete deployment-readiness evidence supplied
Other task constraints	Quality, resources and robustness require their own observations; passing one criterion does not settle them

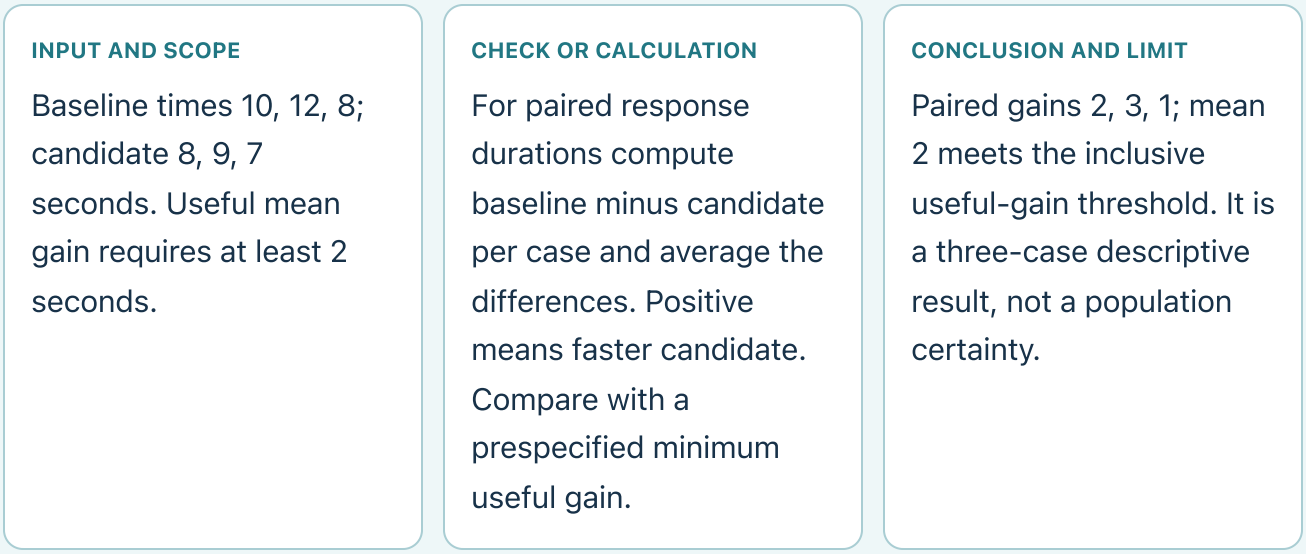
Main paired-gain ledger and checked reasoning

Main paired case	Baseline seconds	Candidate seconds	Gain: baseline – candidate
Q1	10	8	2
Q2	12	9	3
Q3	8	7	1
Mean gain	(2+3+1)/3	2 seconds	Meets inclusive mean threshold 2
Individual variation	Q3 gain is1 second	Mean rule still met	Not a claim every case meets an individual 2-second target
Remaining uncertainty	Only three supplied paired cases	No interval or statistical test supplied	No population-certainty or causal claim

Main gains are 2, 3 and 1 seconds, averaging 2. Equality meets the at-least-two-second mean rule, even though Q3 gains only one second. Here A–B defines time saved, unlike the prior B–A difference. Picture: A has twenty errors per thousand and B fifteen, a reduction of five per thousand, 0.5 percentage points or 25% relative to A. Five misses the required ten.

Changed B with eight errors gives twelve fewer and a 1.2-point reduction, meeting only that criterion. Selected times 6/8/10 versus 5/7/9 give gains 1/1/1 and mean 1, below the new 1.5-second threshold despite positive improvement. No uncertainty, significance, causal or complete-readiness result follows.

Effect size and practical importance — evidence trace



The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Guided: Paired gains 2, 3, 1; mean 2 meets the inclusive useful-gain threshold. It is a three-case descriptive result, not a population certainty. Qualification: A meaningful sample effect may remain statistically uncertain; statistical significance can also concern a practically tiny effect. Exit: accept a concrete variation with its predicted outcome and a stated condition that would challenge the **mechanism**; reject a claim resting on unmeasured **evidence**.

Baseline times 10, 12, 8; candidate 8, 9, 7 seconds. Useful mean gain requires at least 2 seconds.

Paired gains 2, 3, 1; mean 2 meets the inclusive useful-gain threshold. It is a three-case descriptive result, not a population certainty.

Compare the intermediate **evidence** with the final statement. Explain why each number or decision follows from the stated **input**s.

The conclusion belongs to this scenario and procedure; changing the **input**s, evaluation population or authority can change what it establishes. Retain the raw record so a later revision can be compared honestly.

Mean/individual, sign and practical-scope probes

Effect/practicality probe	Expected interpretation
Positive baseline-minus-candidate gain	Candidate is faster under this sign convention
Mean equals the minimum useful gain	Meets an at-least threshold
One case below mean target	Does not automatically fail a rule about the mean
Relative reduction sounds large	Report absolute units and baseline rate too
Observed reduction below practical target	Still a descriptive change, but criterion not met
Observed criterion met	No automatic significance, causal or deployment conclusion
Threshold chosen after inspecting outcomes	Disclose revision rather than claim prespecified importance
Faster response is wrong or over resource budget	Other requirements can still block practical usefulness
Three supplied pairs	Do not invent a population effect interval or certainty

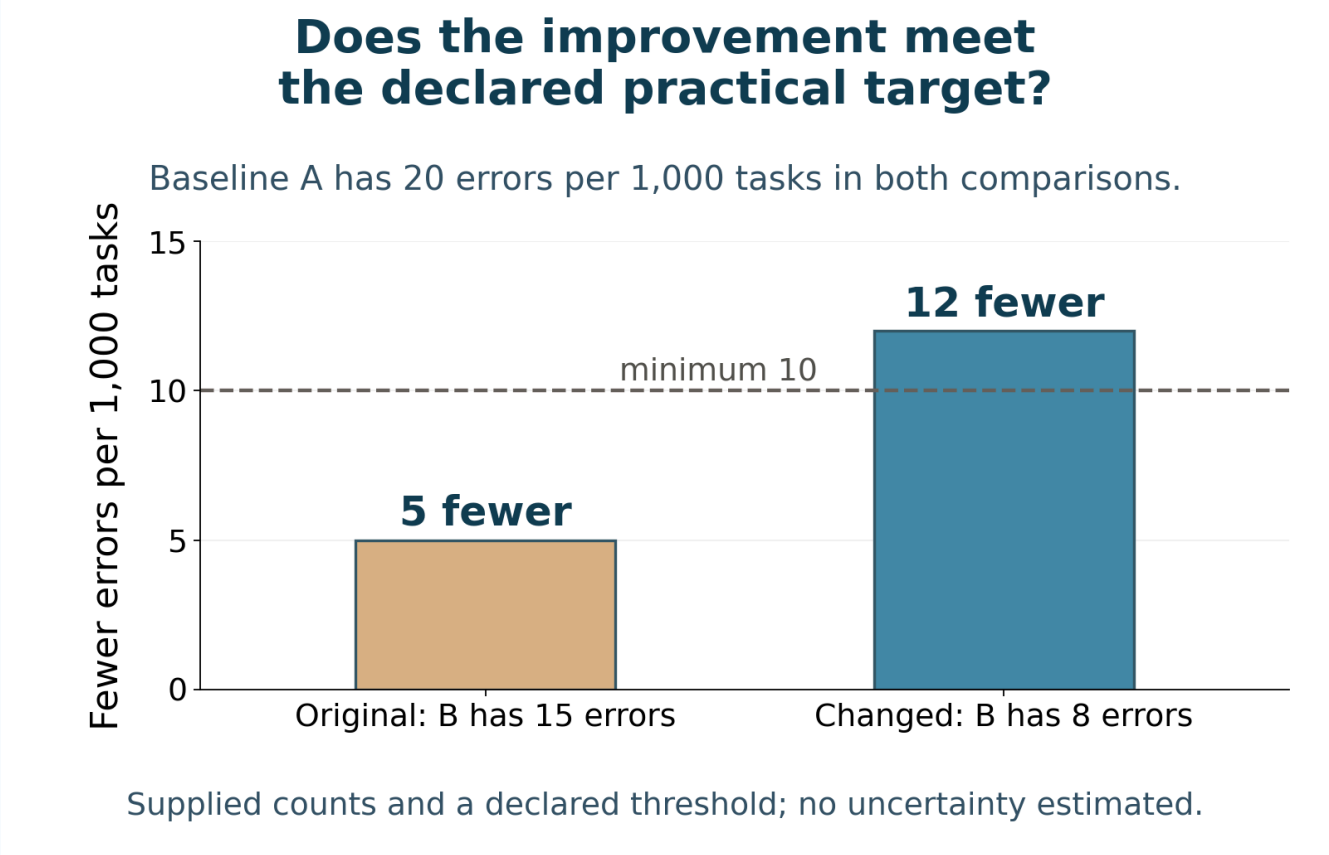
Separate error magnitudes and native threshold comparison

An improvement with a **practical threshold**. A team compares error rates against a predeclared **practical threshold**.

A makes 20 errors in 1000 tasks; B makes 15. Practical improvement requires at least 10 fewer errors per 1000.

The picture is a separate error-rate comparison on one thousand tasks per version. A has 20 errors, or2%; B has 15, or1.5%. The reduction is five errors per thousand, equal to0.5 percentage points and 25% relative to A's error count. The declared practical rule requires at least ten fewer errors per thousand, so this observed reduction misses it.

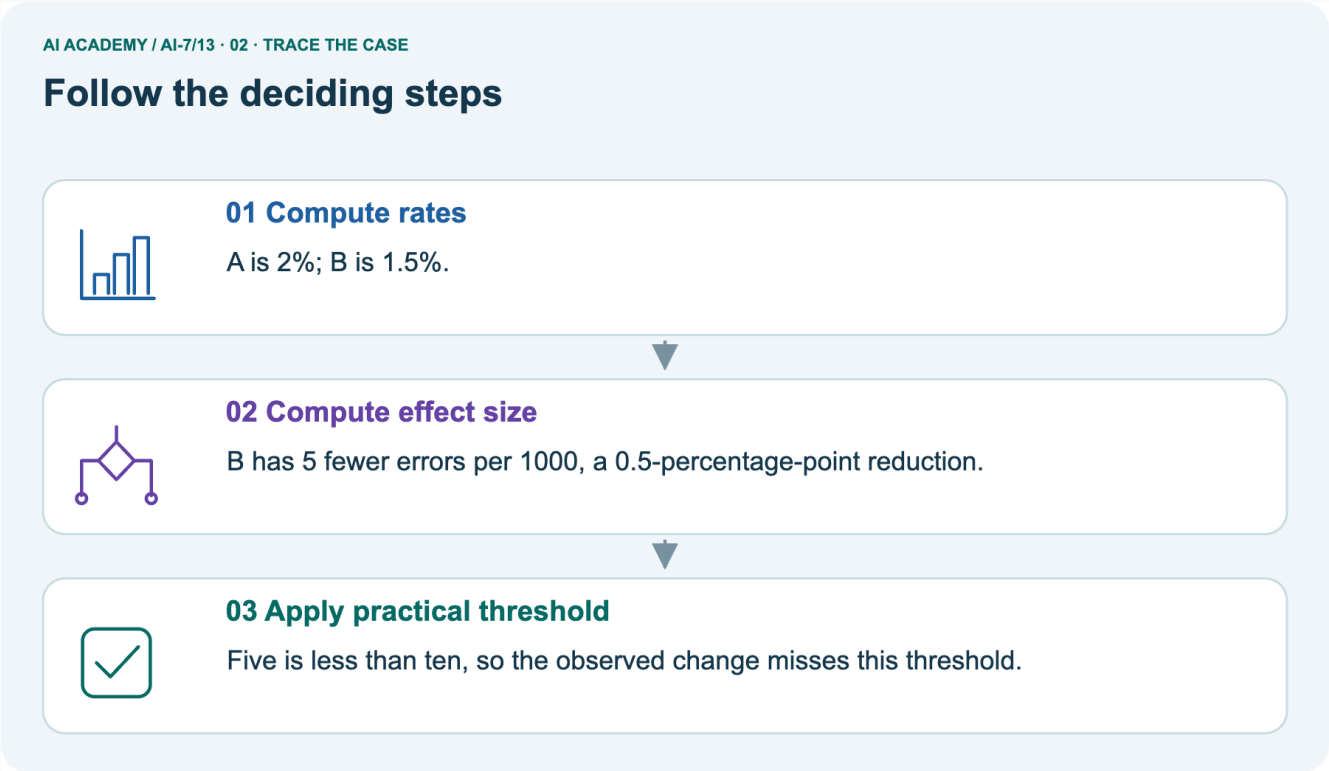
Changed B with eight errors gives twelve fewer, a1.2-percentage-point reduction, meeting that rule. These descriptive counts supply no uncertainty interval or significance result. Meeting this one criterion is not automatic evidence of all quality, resource or deployment requirements.



Separate error comparison	Original B	Changed B
A baseline error count	20/1000 =2%	Same20/1000 =2%
B error count/rate	15/1000 =1.5%	8/1000 =0.8%
Fewer errors per 1000	20−15 =5	20−8 =12
Absolute rate reduction	0.5 percentage points	1.2 percentage points

Separate error comparison	Original B	Changed B
Relative error reduction	5/20 =25%	12/20 =60%
At least 10 fewer required	Misses:5 <10	Meets:12 ≥10
Statistical uncertainty	Not supplied	Still not supplied

Picture trace and changed-error-count key



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Compute rates	A is 2%; B is 1.5%.
2. Compute effect size	B has 5 fewer errors per 1000, a 0.5-percentage-point reduction.
3. Apply practical threshold	Five is less than ten, so the observed change misses this threshold.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Effect size describes how much the outcome changes. Practical importance depends on the real decision criterion; relative changes can sound large while absolute changes remain small.

No uncertainty or statistical significance is supplied; these counts alone cannot establish a general effect.

Change: B makes 8 errors in 1000.

Prediction: The reduction becomes 12 per 1000 and meets the stated **practical threshold**.

Reason: Twenty minus eight exceeds the required ten fewer errors.

▼ Earlier wording, alternatives and picture transcripts

Begin by writing the question the procedure is intended to answer. Identify the information already supplied, the information it will produce and the information still missing.

These are constructed classroom examples; they do not require live accounts, private records or an external service. Do not treat a plausible explanation as a measured outcome.

Before applying the procedure, write its units, denominator or decision boundary where relevant. A partner should be able to repeat each step using the same **input**s.

If a required **input** is absent or violates the stated contract, record that condition explicitly instead of quietly replacing it with a convenient value.

Term	Meaning
Effect size	The magnitude of a difference under a declared scale
Paired difference	A within-unit comparison
Practical threshold	A prespecified magnitude considered useful

Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a **prediction**, then explain the deciding step.

These teacher notes contain solutions; do not reveal the independent case answer during modelling.

Optional local lab: predict the printed result before running this standalone Python 3 example. Change one relevant boundary **input**, record the outcome and state the contract or statistical assumptions.

The program uses fictional values and the Python standard library; it performs no external operations.

```
def paired_gain(baseline, candidate):
    if not baseline or len(baseline) != len(candidate):
        raise ValueError('paired lengths')
    differences = [a - b for (a, b) in zip(baseline, candidate)]
    return (differences, sum(differences) / len(differences))

print(paired_gain([10, 12, 8], [8, 9, 7]))
```

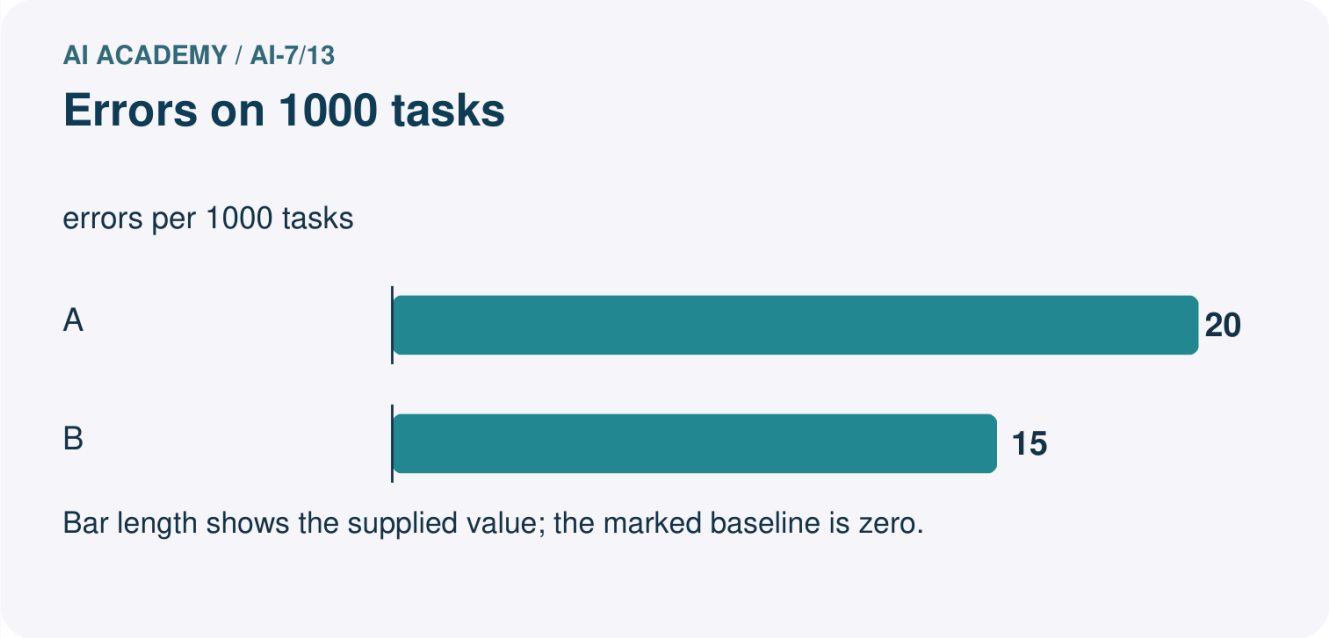
Thinking move	Teacher prompt
Plan	What is given? What is the question? Which procedure fits these facts?

Thinking move	Teacher prompt
Do	Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.
Check	Does the output answer the question? Which assumption or limitation prevents a larger claim?

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

A pictorial worked case: Effect size and practical importance

The facts we will use



Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-7/13

Effect size and practical importance

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Compute rates

A is 2%; B is 1.5%.



2. Compute effect size

B has 5 fewer errors per 1000, a 0.5-percentage-point reduction.



3. Apply practical threshold

Five is less than ten, so the observed change misses this threshold.

CHANGE ONE THING

B makes 8 errors in 1000.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

Change one thing in the pictured case

Ask while showing the picture

Cover the result. Ask learners to name the **input**s and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its **evidence**. Then change the supplied condition and compare.

The picture is a representation; it does not document execution of a real AI model.

MAIN LESSON · SECTION 3

Model a step,then remove help

> Collect each learner’s reasoning before showing a key.

Current completion task and checked reasoning

Completion task: Baseline times 10, 12, 8; candidate 8, 9, 7 seconds. Useful mean gain requires at least 2 seconds. Show the intermediate steps and give the conclusion with its scope.

Teacher check: Teacher explanation: Paired gains 2, 3, 1; mean 2 meets the inclusive useful-gain threshold. It is a three-case descriptive result, not a population certainty.

Start by identifying the given **input**s together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood.

For partners, alternate explainer and checker; collect a response from every learner before discussion.

Individual misconception check

Use the worked case for **Effect size** and practical importance. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Expected correction: A meaningful sample effect may remain statistically uncertain; statistical significance can also concern a practically tiny effect.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class.

If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

Pictured reasoning and changed case

Calculate the absolute error-rate reduction from 20/1000 to 15/1000. Does it meet the ten-fewer-errors threshold, and what changes when B has eight errors?

A is 2%; B is 1.5%. B has 5 fewer errors per 1000, a 0.5-percentage-point reduction. Five is less than ten, so the observed change misses this threshold.

Changed case: The reduction becomes 12 per 1000 and meets the stated **practical threshold**. Twenty minus eight exceeds the required ten fewer errors.

Changed-case reasoning: The reduction becomes 12 per 1000 and meets the stated **practical threshold**. Twenty minus eight exceeds the required ten fewer errors.

Collect each learner's explanation before discussion. Use the first missing connection between **input**, deciding step and result to choose support. This question is worked-case practice, not fresh mastery **evidence**.

Accessible support and extension

Use the worked example in Sections 3 and 6. Collect a **prediction** before revealing its result; ask the learner to point to each intermediate fact.

Provide an **input**/procedure/result/limit frame and a calculator when needed. Ask students to identify the first step before completing the whole trace.

Extension: compare two concrete variants of this **mechanism** while holding unrelated conditions fixed; report both successful and unsuccessful outcomes.

▼ Earlier wording, alternatives and picture transcripts

Time or stage	Teacher action
Session 1: 0–7 minutes	Retrieve the prerequisite; identify the input, procedure and evidence source.
7–18 minutes	Explain See how it works and Words that unlock the idea. Complete W1 with a mechanism-based explanation.
18–32 minutes	Trace the declared procedure and worked case. Complete W2, recording intermediate steps.
32–45 minutes	Discuss the diagram with the worked input; use W3 to distinguish measured claims and limitations.
Session 2: 0–8 minutes	Retrieve the worked reasoning and collect the guided response before discussing it.
8–20 minutes	Compare a proposed boundary variant with the original case; identify what remains fixed.
20–35 minutes	Collect W4 independently. Require the procedure trace and the bounded conclusion.
35–45 minutes	Use W5 and exit responses for feedback. Administer the fresh retry after feedback and preserve both attempts.

Reduce help deliberately

Individual reasoning check

Pictorial practice question

Reasoned teacher answer

MAIN LESSON · SECTION 4

Run the complete supplied investigation

> Separate predictions,actual runs and paper traces.

Complete inputs,expected results and limits

Main gains 2, 3, 1 average 2 and meet the at-least-two-second mean rule, which does not require every individual gain to equal two. Picture five fewer errors misses ten, while changed twelve fewer meets it; no uncertainty or significance supplied. Selected baseline 6, 8, 10 versus candidate 5, 7, 9 yields gains 1, 1, 1 and mean 1, below 1.5 seconds despite positive improvement.

Preserve sign, units and the declared criterion without inferring a complete practical or statistical result.

Diagnose the current misconceptions

The worked result establishes every future case: A meaningful sample effect may remain statistically uncertain; statistical significance can also concern a practically tiny effect.

A missing **input** may be replaced without recording the change: A replacement changes the **evidence** or contract. Record the change and its justification; do not silently turn unknown into a measured value.

Repeating an assessed case is new independent **evidence**: A replay checks consistency. An example used for feedback or revision cannot later be described as untouched final evaluation.

Use Section 9 for the **evidence** ledger. Add a boundary or missing-**input** case and record the expected response before checking. More successful repeats of one case do not establish wider coverage.

Choose accessible support

What the response shows	Next teaching action
Missing or misread facts	Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.
Wrong procedure	Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.
Arithmetic or implementation error	Trace intermediate values or lines; preserve the first mismatch and retry that step.

What the response shows	Next teaching action
Correct result without a reason	Ask for a trace and a changed case before marking independent understanding.
An unsupported wider claim	Ask which supplied evidence supports the claim and which further observation would be needed.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal.

Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

▼ **Earlier wording, alternatives and picture transcripts**

Match the support to the first error

MAIN LESSON · SECTION 5

Assess the new research scenario

> Keep the selected answer private until after a first attempt.

Selected independent task and checked answer

Selected case: Baseline times are 6, 8, 10 seconds; candidate times are 5, 7, 9 seconds on the same three cases in the same order. Define each gain as baseline minus candidate. Useful mean gain must be at least 1.5 seconds. Calculate and decide.

Teacher explanation: Gains 1, 1, 1; mean 1 fails 1.5 threshold despite positive sample improvement.

Judge explanation,trace and limits

Dimension	Evidence required
Explanation	Names the applicable idea or procedure and ties it to the supplied facts.
Trace and result	Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.
Limits	States the assumptions, missing evidence or conditions under which the result would change.

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace.

Do not penalise an honestly recorded model failure if the learner correctly explains its **evidence** and limits.

Give one usable repair

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case.

Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

W1 Explain the mechanism

Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

Teacher explanation: **Effect size** describes magnitude on a stated scale. Practical importance depends on task requirements, not merely a nonzero difference.

For paired response durations compute baseline minus candidate per case and average the differences. Positive means faster candidate. Compare with a prespecified minimum useful gain.

W2 Trace the worked case

Baseline times 10, 12, 8; candidate 8, 9, 7 seconds. Useful mean gain requires at least 2 seconds. Show the intermediate steps and give the conclusion with its scope.

Teacher explanation: Paired gains 2, 3, 1; mean 2 meets the inclusive useful-gain threshold. It is a three-case descriptive result, not a population certainty.

W3 Audit the limit

A report omits this qualification: A meaningful sample effect may remain statistically uncertain; statistical significance can also concern a practically tiny effect.

Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

Teacher explanation: A meaningful sample effect may remain statistically uncertain; statistical significance can also concern a practically tiny effect.

State the observed result separately from the unverified or out-of-scope claim; preserve the original **evidence** rather than inventing a missing fact.

W5 Design a revealing follow-up

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

Do not reuse W4 as a new final test after receiving feedback.

Teacher explanation: Accept a concrete variant of the worked **input** or a missing/conflicting/boundary condition, with an expectation derived from the declared procedure.

Require preserved settings, original **output**s and data roles. A meaningful sample effect may remain statistically uncertain; statistical significance can also concern a practically tiny effect.

The new case should test this particular **mechanism** rather than merely restate the original result.

▼ Earlier wording, alternatives and picture transcripts

Guided and exit questions

Guided: Paired gains 2, 3, 1; mean 2 meets the inclusive useful-gain threshold. It is a three-case descriptive result, not a population certainty. Qualification: A meaningful sample effect may remain statistically uncertain; statistical significance can also concern a practically tiny effect. Exit: accept a concrete variation with its predicted outcome and a stated condition that would challenge the **mechanism**; reject a claim resting on unmeasured **evidence**.

W1 Explain the mechanism

W2 Trace the worked case

W3 Audit the limit

W4 Apply a new case independently

Baseline times are 6, 8, 10 seconds; candidate times are 5, 7, 9 seconds on the same three cases in the same order. Define each gain as baseline minus candidate. Useful mean gain must be at least 1.5 seconds. Calculate and decide.

Teacher explanation: Gains 1, 1, 1; mean 1 fails 1.5 threshold despite positive sample improvement.

W5 Design a revealing follow-up

Independent assessment and fresh retry

W4: Gains 1, 1, 1; mean 1 fails 1.5 threshold despite positive sample improvement. Require intermediate reasoning and the stated scope. Fresh retry: Baseline5, 7, 9; candidate3, 5, 8; threshold1.5. Expected reasoning: Gains2, 2, 1; mean5/3≈1.667 meets threshold on three cases. If students need support, record it and reassess with another new case rather than awarding independence to a copied answer.

Assess independent transfer

Selected case answer

Feedback that the learner uses

MAIN LESSON · SECTION 6

Connect, support and revisit research reasoning

> Use saved evidence and assumptions to choose the next question.

Project and next-week bridge

Add a labelled **effect size** and practical importance **evidence** sheet to Reproducible AI Research Study. Show how you can explain the **mechanism** and terminology of **effect size** and practical importance.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

Provide an **input**/procedure/result/limit frame and a calculator when needed. Ask students to identify the first step before completing the whole trace.

Extension: compare two concrete variants of this **mechanism** while holding unrelated conditions fixed; report both successful and unsuccessful outcomes.

Next connection: Statistical tests and assumptions. Preserve the current case record and retrieve its **evidence** boundary before moving on.

Retain the project ledger and assess a changed case after support. Check the relevant prerequisite before carrying this component forward.

For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

Delayed return and support record

Suggested return points in this level: ai-7/14, ai-7/16, ai-7/19. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used.

For a new level, revisit the previous capstone before checking the new prerequisites.

Record	Teacher evidence
Learner code and date	
Concept and case used; prior exposure	
Explanation / trace / limits	
Support and first repair	
Changed case and delayed outcome	
Next teaching action	

Use the **evidence** to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

▼ Earlier wording, alternatives and picture transcripts

Plan the delayed return

Project evidence from this case

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Alternative planning checklist**

REFERENCE · SECTION 7

Reference · Alternative planning checklist

> Use alongside the main route when helpful.

Planning,assessment and project reminders

Teaching target: Explain the **mechanism** and terminology of **effect size** and practical importance

Prepare the labelled worked-case picture, the matching workbook, fictional **input** cards and a place for individual responses. Check a local Python 3 environment before class; have a paper-trace version ready.

Session · Plan · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Plan:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Use Your mission, Notice the evidence, See how it works, Words that unlock the idea and Follow a worked example; record the brief recall in the workbook.

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>
> **Resources:** Use Run the investigation, Build your understanding, Connect your project and Show what you know. Reserve further time for the full task set. Use a code example only where one is actually supplied.

Before class, solve the supported practice: Baseline times 10, 12, 8; candidate 8, 9, 7 seconds. Useful mean gain requires at least 2 seconds. Show the intermediate steps and give the conclusion with its scope.

Reasoned practice answer: Teacher explanation: Paired gains 2, 3, 1; mean 2 meets the inclusive useful-gain threshold. It is a three-case descriptive result, not a population certainty.

Check every learner before discussion: Use the worked case for **Effect size** and practical importance. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Misconception repair: A meaningful sample effect may remain statistically uncertain; statistical significance can also concern a practically tiny effect.

Support: read the exact case aloud, enlarge the labelled picture, supply **input** cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

Extension: B makes 8 errors in 1000. Require a **prediction** and a reason before comparing. Expected result: The reduction becomes 12 per 1000 and meets the stated **practical threshold**.

Reason: Twenty minus eight exceeds the required ten fewer errors.

Independent assessment: withhold the reserved answer until the learner has attempted the new case. Record support separately from correctness.

Criterion	Not yet	With support	Independent
Explanation	Names an answer without a deciding fact	Uses a hint to link fact and decision	Explains the deciding fact in own words
Trace or test	Cannot connect input and result	Completes steps with prompts	Traces or tests a changed case accurately
Limits and revision	Claims more than evidence supports	Names a limit after a prompt	States a defensible limit and useful next test

Project checkpoint: Test changed cases. Keep normal, boundary and failure cases; compare expected and observed results.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and **evidence**.

▼ **Reference · Feedback,labs and repair paths**

REFERENCE · SECTION 8

Reference · Feedback,labs and repair paths

> Retain the first attempt and the evidence used for a repair.

Feedback and record guidance

Use the two formative questions as supported practice, not as independent mastery scores. Discuss the deciding **evidence** rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

Ask students to predict before running code. Preserve errors and actual **output**s; collect a changed test and an explanation. The branch-boundary and indexing lessons deliberately include failure cases.

Use the earlier worked-explanation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

Collect the completed-work portfolio as **evidence**. Record support separately from correctness, and assess explanation, trace/test and limitations. Progress ticks and quiz selections alone do not establish mastery.

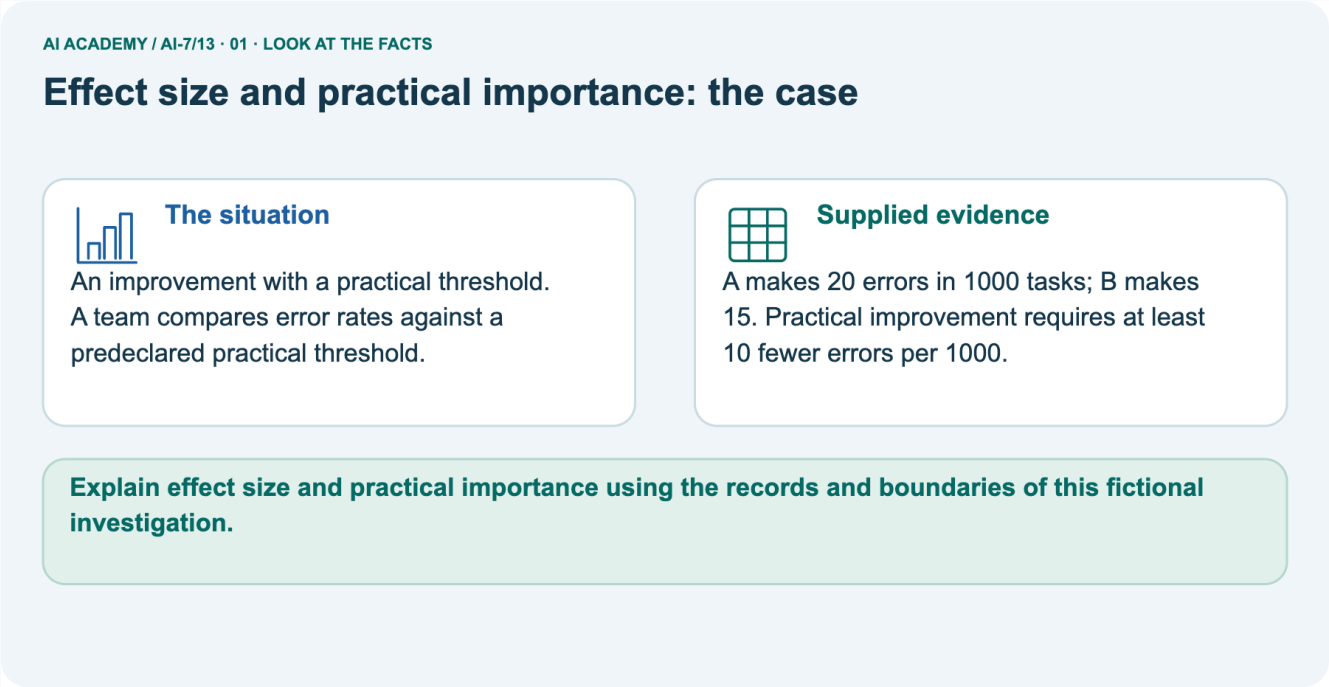
▼ **Reference · Teach with the six pictures**

REFERENCE · SECTION 9

Reference · Teach with the six pictures

> Choose one picture for a particular reasoning move.

01. Effect size and practical importance: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

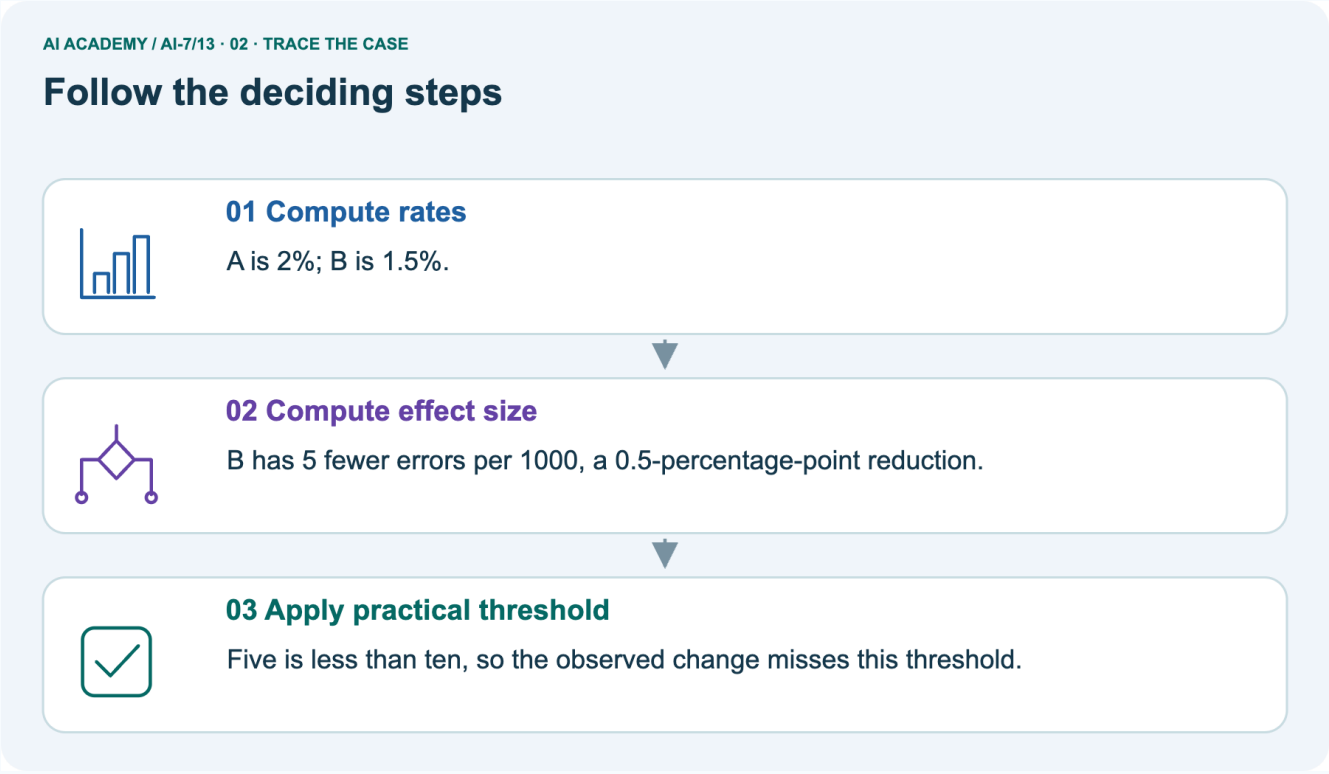
An improvement with a **practical threshold**. A team compares error rates against a predeclared **practical threshold**. Supplied facts: A makes 20 errors in 1000 tasks; B makes 15.

Practical improvement requires at least 10 fewer errors per 1000.

Point to the given fact that matters most. Explain why it matters.

Teacher prompt: Point to the given fact that matters most. Explain why it matters. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

Compute rates: A is 2%; B is 1.5%. → Compute **effect size**: B has 5 fewer errors per 1000, a 0.5-percentage-point reduction. → Apply **practical threshold**: Five is less than ten, so the observed change misses this threshold.

Point to each numbered stage and explain which recorded **input** it uses.

Teacher prompt: Point to each numbered stage and explain which recorded **input** it uses. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

03. Errors on 1000 tasks



Follow the labelled picture with the matching explanation. Open the vector diagram

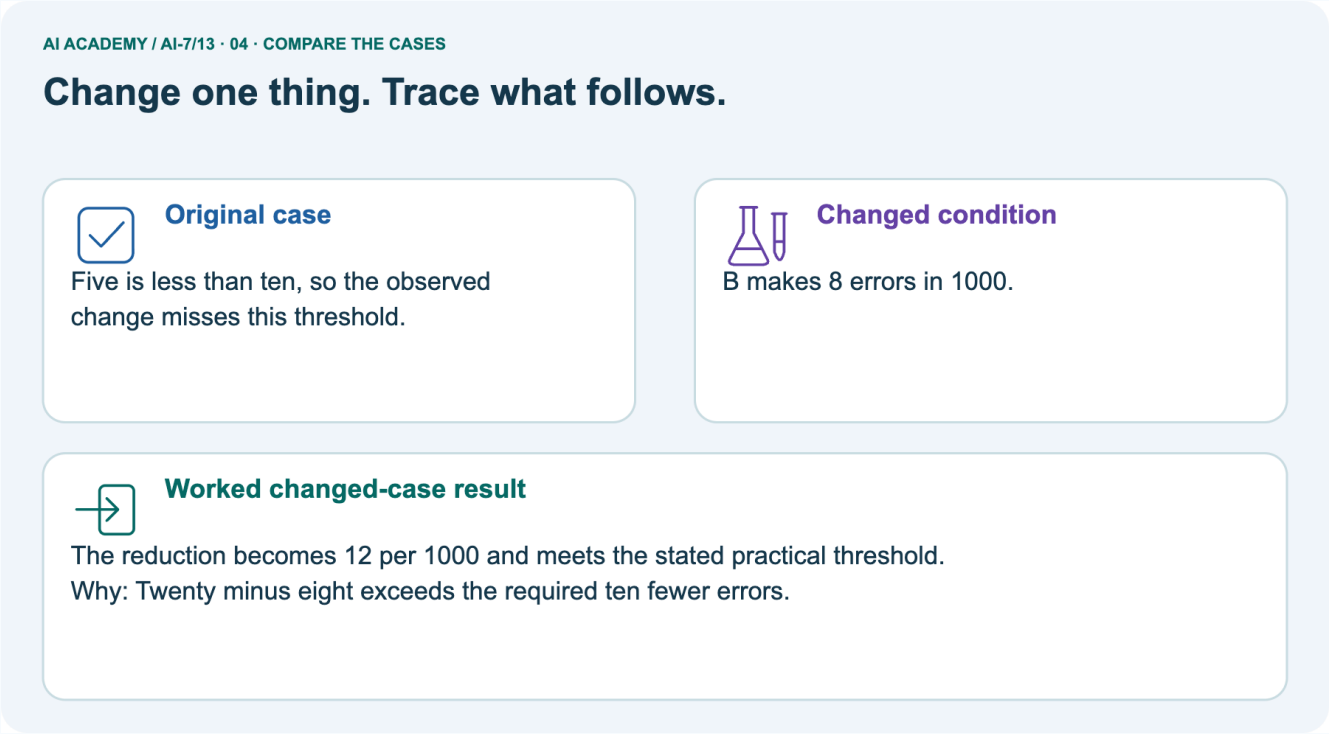
Errors on 1000 tasks. Units: errors per 1000 tasks. A: 20. B: 15.

Read the labels and values. Explain the relationship the model shows.

Teacher prompt: Read the labels and values. Explain the relationship the model shows. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Check row/column meanings, units and denominators before discussing a difference. Supplied toy values are not real class or model results.

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Five is less than ten, so the observed change misses this threshold. Changed condition: B makes 8 errors in 1000. Expected result: The reduction becomes 12 per 1000 and meets the stated **practical threshold**.

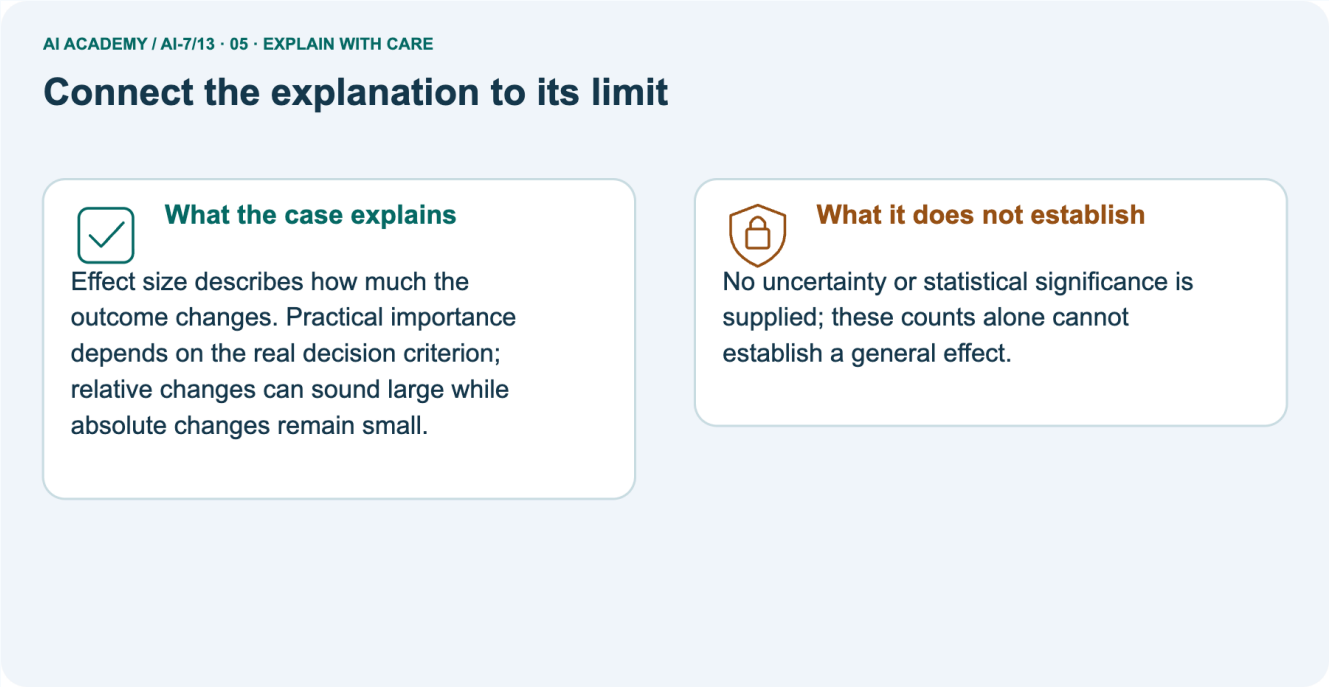
Reason: Twenty minus eight exceeds the required ten fewer errors.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

Teacher prompt: Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude. Ask learners to describe one label or arrow in ordinary words.

Use the adjacent transcript if the image is difficult to see. Changed-case reasoning: Twenty minus eight exceeds the required ten fewer errors.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

Effect size describes how much the outcome changes. Practical importance depends on the real decision criterion; relative changes can sound large while absolute changes remain small.

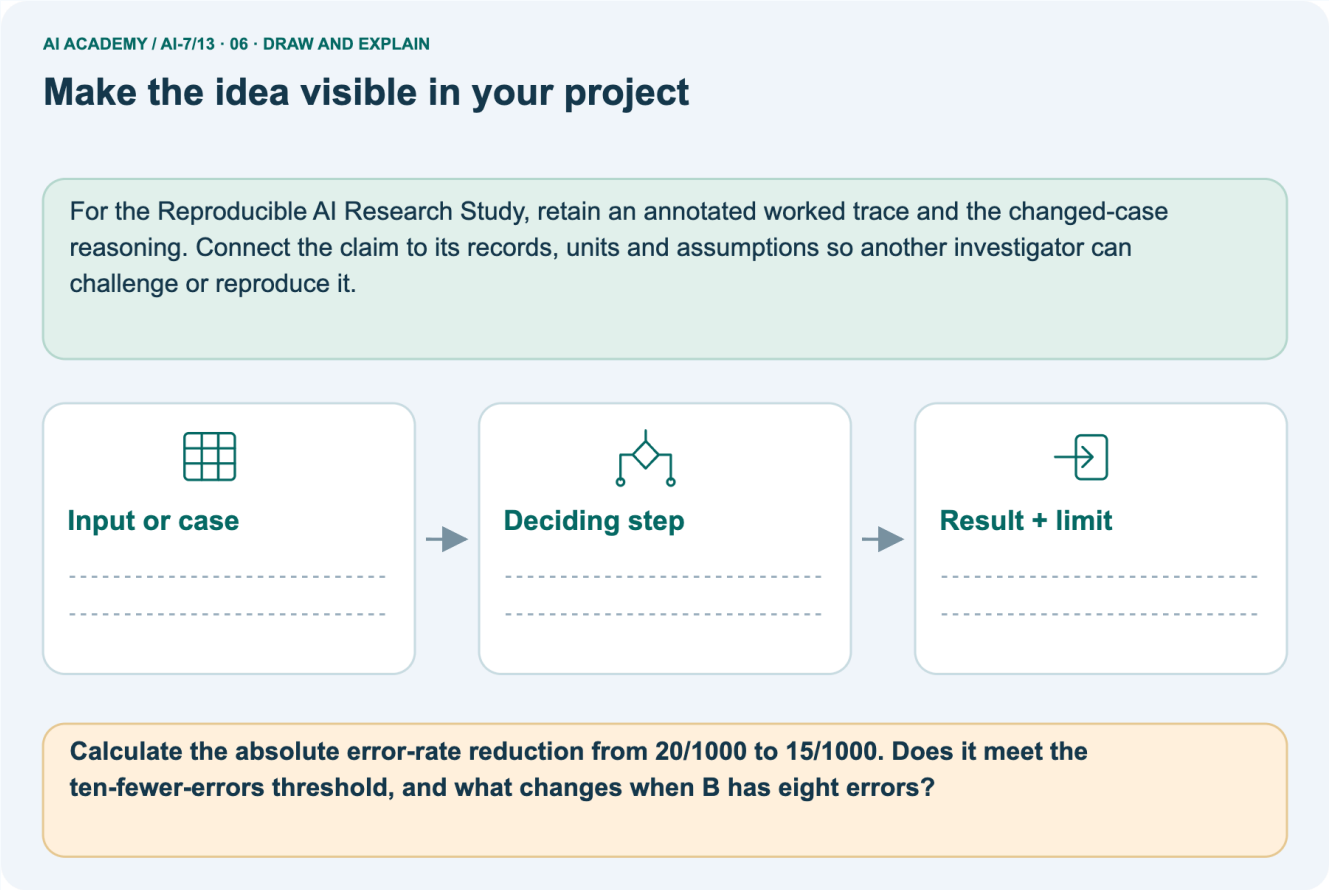
Limit: No uncertainty or statistical significance is supplied; these counts alone cannot establish a general effect.

Explain the case, then point to a limit on the claim.

Teacher prompt: Explain the case, then point to a limit on the claim. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Keep the stated scope: No uncertainty or statistical significance is supplied; these counts alone cannot establish a general effect.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the claim to its records, units and assumptions so another investigator can challenge or reproduce it.

Draw your **input**, deciding step, result and limit. Supported question: Calculate the absolute error-rate reduction from 20/1000 to 15/1000. Does it meet the ten-fewer-errors threshold, and what changes when B has eight errors?

Draw your project trace and justify the deciding step using **evidence**.

Teacher prompt: Draw your project trace and justify the deciding step using **evidence**. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Supported practice key: A is 2%; B is 1.5%. B has 5 fewer errors per 1000, a 0.5-percentage-point reduction. Five is less than ten, so the observed change misses this threshold.

Changed case: The reduction becomes 12 per 1000 and meets the stated **practical threshold**. Twenty minus eight exceeds the required ten fewer errors.

▼ Earlier wording, alternatives and picture transcripts

Use one picture at a time. Hide a worked result before asking for a **prediction**, then reveal it for feedback. These are supported teaching cases; retain the separate independent assessment for a fresh first attempt.

01. Effect size and practical importance: the case

02. Follow the deciding steps

03. Errors on 1000 tasks

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.